



Basic Linux Networking Commands



This [session](#) (L6 S11) will cover essential networking commands that are critical for troubleshooting and managing network-related issues.

Basic networking commands:

Linux provides a variety of networking commands to manage and troubleshoot network connections.

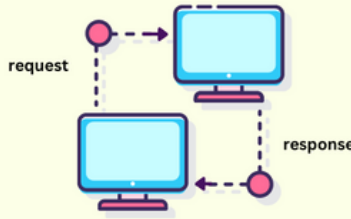
Here are some basic and commonly used commands:

- a) ping
- b) ip
- c) ifconfig
- d) netstat
- e) ss

Command	Description
ping	Test connectivity to a remote host
ifconfig	Configure network interfaces (deprecated)
ip	Manage network interfaces, IP addresses, and routes
netstat	Display network connections and statistics (deprecated)
ss	Display socket statistics and network connections



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Ping Command:

a) ping

- The ping command is used to test connectivity between two nodes on a network.
- It works by sending ICMP echo request packets and waiting for a reply.

Syntax:

ping [options] destination

Examples:

- **Ping a hostname**

ping www.google.com

- **Ping an IP address**

ping 8.8.8.8

- **Send a specific number of packets**

ping -c 4 www.google.com

- **Specify packet size**

ping -s 56 www.google.com



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Linux ip Command with Examples



b) ip

- The ip command manages various network-related configurations such as IP addresses, interfaces, and routing tables.

Syntax:

`ip [object] [command] [options]`

Examples:

- **Display all network interfaces and their IP addresses**

`ip addr show`

- **Display detailed information about a specific interface**

`ip addr show eth0`

- **Assign an IP address to an interface**

`sudo ip addr add 192.168.1.100/24 dev eth0`

- **Remove an IP address from an interface**

`sudo ip addr del 192.168.1.100/24 dev eth0`



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- **Bring an interface up**

```
sudo ip link set eth0 up
```

- **Bring an interface down**

```
sudo ip link set eth0 down
```

- **Display the routing table**

```
ip route show
```

- **Add a static route**

```
sudo ip route add 192.168.2.0/24 via 192.168.1.1
```

- **Delete a static route**

```
sudo ip route del 192.168.2.0/24
```

```
[tecmint@tecmint]$ sudo ip addr flush eth2
[tecmint@tecmint]$ ip -j -p addr show eth2
[ {
    "ifindex": 13,
    "link_index": 14,
    "ifname": "eth2",
    "flags": [ "BROADCAST", "MULTICAST", "UP", "LOWER_UP" ],
    "mtu": 1500,
    "qdisc": "noqueue",
    "operstate": "UP",
    "group": "default",
    "link_type": "ether",
    "address": "02:42:ac:13:01:02",
    "broadcast": "ff:ff:ff:ff:ff:ff",
    "link_netnsid": 0,
    "addr_info": [ ]
  } ]
[tecmint@tecmint]$
```




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ifconfig Command in Linux with Examples



c) ifconfig

- It's command is used to configure network interfaces.
- But it is considered deprecated and replaced by the ip command in modern Linux distributions.

Syntax:

```
ifconfig [interface] [options]
```

Examples:

- **Display all network interfaces**

```
ifconfig
```

- **Display information about a specific interface**

```
ifconfig eth0
```

- **Assign an IP address to an interface**

```
sudo ifconfig eth0 192.168.1.100
```

- **Bring an interface up**

```
sudo ifconfig eth0 up
```

- **Bring an interface down**

```
sudo ifconfig eth0 down
```



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d) netstat

- The netstat command provides information about network connections, routing tables, and interface statistics.
- It is considered **deprecated and replaced by the ss** command in modern Linux distributions.

Syntax:

`netstat [options]`

Examples:

- **Display all active connections**

`netstat`

- **Display all listening ports**

`netstat -l`

- **Display TCP connections**

`netstat -t`

- **Display UDP connections**

`netstat -u`

- **Display routing table**

`netstat -r`

- **Display network interface statistics**

`netstat -i`



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e) ss

- The ss command is used for displaying socket statistics and is a modern replacement for netstat.
- The ss command is used to display socket statistics, and it's a modern replacement for netstat.

Syntax:

ss [options]

Examples:

- **Display all open network connections**

ss -a

- **Display all listening ports**

ss -l

- **Display all TCP connections**

ss -t

- **Display all UDP connections**

ss -U

- **Display network statistics**

ss -s

- **Display processes using network connections**

ss -p